

PRACTICAL WEEK1

TASK1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user. Accept a Linux command as input. Create a child process using `fork()`. Execute the command in the child process using an appropriate `exec()` system call. Allow the parent process to wait for the child using `wait()`. Display the Process ID (PID) of both the parent and child processes.

```
keerthi_priya@LAPTOP-OSGV x + v
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd skill1
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ nano command.c
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ gcc command.c -o command
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ ./command
Enter a Linux command: ls

Child Process
Child PID: 581
command command.c command.c.save

Parent Process
Parent PID: 580
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ |
```

```
GNU nano 8.7.1 cor
{
    printf("\nChild Process\n");
    printf("Child PID: %d\n", getpid());

    execlp(command, command, NULL);

    printf("Command execution failed.\n");
}

else
{
    wait(NULL);

    printf("\nParent Process\n");
    printf("Parent PID: %d\n", getpid());
}

return 0;
}|
```

TASK2: Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the operating system abstracts the following resources: CPU, Memory, Storage and Input/Output (I/O) devices

```

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keerthi_priya@LAPTOP-OSGVCE7A:~$ cd skill1
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ nano command.c
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ gcc command.c -o command
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ ./command
Enter a Linux command: ls

Child Process
Child PID: 581
command command.c command.c.save

Parent Process
Parent PID: 580
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ uname
Linux
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ uname -a
Linux LAPTOP-OSGVCE7A 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          46 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 18
  On-line CPU(s) list:  0-17
Vendor ID:              GenuineIntel
Model name:              Intel(R) Core(TM) Ultra 5 125H
CPU family:              6
Model:                  170
Thread(s) per core:     2
Core(s) per socket:     9
Socket(s):               1
Stepping:                4
BogoMIPS:                5990.39
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                        r pge mca cmov pat pse36 clflush mmx fxsr sse s
                        se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                        _tsc rep_good nopl xtopology tsc_reliable nonst
                        op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                        sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                        opcnt tsc_deadline_timer aes xsave avx f16c rdr
                        and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                        brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi

```

```

keerthi_priya@LAPTOP-OSGV X + v
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-17
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:         Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled v
                        ia prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user
                        pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB con
                        ditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_
                        S
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M 1 disk
sdb   8:16   0 159.4M 1 disk
sdc   8:32   0 2G 0 disk [SWAP]
sdd   8:48   0 1T 0 disk /mnt/wslg/distro
/
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ ps
  PID TTY          TIME CMD
  513 pts/0    00:00:00 bash
  710 pts/0    00:00:00 ps
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ top
top - 06:41:40 up 12 min, 1 user, load average: 0.02, 0.03, 0.00
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie

```

```
keerthi_priya@LAPTOP-OSGV x + v
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ ps
  PID TTY          TIME CMD
   513 pts/0    00:00:00 bash
   710 pts/0    00:00:00 ps
keerthi_priya@LAPTOP-OSGVCE7A:~/skill1$ top
top - 06:41:52 up 12 min,  1 user,  load average: 0.01, 0.03, 0.00
Tasks:  24 total,   1 running, 23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  7651.2 total,   7013.2 free,   608.1 used,   181.4 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7043.1 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24552   15536   11596 S   0.0   0.2   0:00.82 systemd
    2 root        20   0    3180    2212    2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   46 root        19  -1   50400   16732   15456 S   0.0   0.2   0:00.21 systemd-journal
   81 systemd+    20   0   22416   14512   12012 S   0.0   0.2   0:00.06 systemd-resolve
   90 root        20   0   35444   12388    9220 S   0.0   0.2   0:00.46 systemd-udevd
  107 root        20   0    2888    1952    1820 S   0.0   0.0   0:00.04 chronyd-starter
  113 root        20   0    4472    3136    2888 S   0.0   0.0   0:00.00 cron
  122 message+    20   0    8796    5324    4636 S   0.0   0.1   0:00.05 dbus-daemon
  165 root        20   0   44096   29864   14724 S   0.0   0.4   0:00.26 networkd-dispat
  172 root        20   0    18492   9436    8140 S   0.0   0.1   0:00.08 systemd-logind
  185 syslog      20   0   220548   5912    4648 S   0.0   0.1   0:00.07 rsyslogd
  224 root        20   0    5492    2732    2548 S   0.0   0.0   0:00.00 agetty
  247 _chrony      20   0   23820   11328    7200 S   0.0   0.1   0:00.10 chronyd
  260 _chrony      20   0   12096    2468    1824 S   0.0   0.0   0:00.01 chronyd
  269 root        20   0   123460   32612   17996 S   0.0   0.4   0:00.23 unattended-upgr
  346 root        20   0    8648    5512    4668 S   0.0   0.1   0:00.01 login
  391 keerthi+    20   0   22512   13412   10956 S   0.0   0.2   0:00.14 systemd
  393 keerthi+    20   0   22920    4116    2112 S   0.0   0.1   0:00.00 (sd-pam)
  424 keerthi+    20   0    6136    5412    3856 S   0.0   0.1   0:00.04 bash
  508 root        20   0    3184    1124     980 S   0.0   0.0   0:00.00 SessionLeader
  509 root        20   0    3200    1260    1108 S   0.0   0.0   0:00.03 Relay(513)
  513 keerthi+    20   0    6268    5536    3864 S   0.0   0.1   0:00.10 bash
  720 keerthi+    20   0    8172    6008    3900 R   0.0   0.1   0:00.02 top
```